

SHRI G. S. INSTITUTE OF TECHNOLOGY AND SCIENCE
I MID TERM TEST FEBRUARY 2017
B.E IST YEAR CH-10502 CHEMISTRY

TIME: 1 HOUR

MM:15

NOTE: Attempt any Three Questions

~~Q.1~~ Giving suitable examples, discuss addition and condensation polymerization. 5

OR

Discuss Thick and Thin film mechanisms of lubrication.

~~Q.2~~ Calculate the amount of lime and soda required for softening 1,00,000 litres of water using 139 ppm of $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$ as a coagulant. The results of the analyses of raw and softened water are as follows: 5

Analysis of raw water: $\text{Ca}^{+2} = 160\text{ppm}$; $\text{Mg}^{+2} = 288\text{ppm}$; $\text{HCO}_3^- = 1464\text{ppm}$; dissolved $\text{CO}_2 = 20\text{ppm}$.

Analysis of treated water: $\text{CO}_3^{--} = 60\text{ppm}$; $\text{OH}^- = 34\text{ppm}$. 47.7 180 2633

~~Q.3~~ Write short notes on: 5
(a) Aniline point and Extreme pressure lubricants.

OR

(b) Kevlar and Bakelite

~~Q.4~~ What do you understand by Hardness of water? Distinguish between temporary and permanent hardness giving suitable example, also enlist external and internal treatment methods required for water softening. 5

~~Q.5~~ Give a detailed account of Electrochemical theory of Wet Corrosion giving its importance. 5

SHRI G.S. INSTITUTE OF TECHNOLOGY AND SCIENCE, INDORE

CLASS TEST II, MARCH 2017

B.E. I YEAR

CH10502 (CHEMISTRY)

1. Time: 01 Hr.

M. Marks: 15

Note: ATTEMPT ANY THREE QUESTIONS

Q.1 Define Viscosity and Viscosity index. An unknown oil sample has a Say bolt Universal Viscosity (S.U.V.) of 58 seconds at 210 °F and 580 seconds at 100 °F. The high Viscosity Index (V.I.) Standard (Pennsylvanian oil) has S.U.V. OF 58 seconds at 210 °F and 430 seconds at 100 °F. The low V.I. Standard (Gulf oil) has a S.U.V. Of 58 seconds at 210 °F and 780 seconds at 100 °F. Viscosity Index of unknown oil is: 5

Q.2 Write short note (ANY TWO): 5

- a) Synthetic Lubricant
- b) Proximate analysis
- c) Semi Solid Lubricant

Q.3 •What do you mean by refractory? Describe Seger cone or RUL test of refractory. 5

Q.4 Discuss manufacturing process of Portland Cement. 5

Or

What do you mean by Refectory? Describe Segar Cone Test or RUL Test? 5

Q.5 Discuss Preparation Properties & Uses of (ANY TWO): 5

- a) Nylon66
- b) Polyester
- c) Teflon

Q.6 Draw labeled diagram of Bomb Calorimeter and derive the formula for calorific value employed in this. 5

SHRI G.S. INSTITUTE OF TECHNOLOGY & SCIENCE, INDORE
CLASS TEST I, SEPTEMBER 2016

B.E. I YEAR

PAPER: CHEMISTRY (CH10502)

Max. Marks: 15

Max. Time: 01 Hr.

Note: Attempt any three questions. All questions carry equal marks.

- ~~Q.1~~ a) Define (i) Polymer (ii) Degree of polymerization 2 + 3
b) Discuss any three types of classifications of polymers.
- Q.2 Define various types of calorific value. Give labeled diagram of Bomb calorimeter. Describe briefly various types of corrections employed for Calorific value determined by Bomb calorimeter. 5
- ~~Q.3~~ a) Discuss hardness of water under following heads: Definition, Types and Reasons. 2.5 + 2.5
b) A water sample on chemical analysis gave the following concentration of different salts:
 $\text{Ca}(\text{HCO}_3)_2 = 16.2 \text{ mg/L}; \quad \text{Mg}(\text{HCO}_3)_2 = 0.0073 \text{ g/L};$
 $\text{MgCl}_2 = 9.5 \text{ mg/L}; \quad \text{CaSO}_4 = 13.6 \text{ mg/L}.$
Calculate the temporary, permanent and total hardness in mg/L, ppm, $^\circ\text{Cl}$, French. 3 + 2
- Q.4 Write short notes:
(a) Proximate analysis of coal (b) Pulverization of coal 1.5 + 1.5 + 2
- ~~Q.5~~ Discuss following boiler troubles:
(a) Scales and sludges
(b) Boiler corrosion
(c) Caustic embrittlement 2 + 2 + 1
- Q.6 Discuss structure, property and uses of the followings:
(a) Kevlar (b) Polyester (c) Polythene

B.E. I YEAR
CLASS TEST I, September 2015
Paper: CH1058 (Chemistry of Engineering Materials & Processes)

M. Marks 20

Time: 01 Hr.

Note: Attempt any Four Questions. All Questions Carry Equal Mark.

- Q.1 Define Hardness, types of hardness. Calculate the temporary and permanent hardness of a sample of water containing 5
- | | |
|--------------------------------|------------|
| a) $\text{Ca}(\text{HCO}_3)_2$ | = 162 ppm; |
| b) $\text{Mg}(\text{HCO}_3)_2$ | = 146 ppm; |
| c) MgSO_4 | = 120 ppm; |
| d) CaSO_4 | = 136 ppm; |
| e) CaCl_2 | = 111 ppm; |
| f) MgCl_2 | = 95 ppm. |
- Q.2 Discuss Addition and Condensation methods of Polymerisation. 5
- Q.3 Explain Thin film and Thick film mechanisms of lubrication with suitable examples. 5
- Q.4 Write structure, properties and uses of the following: [Any Two] 5
- (a) Nylon 6:6 (b) Kevlar (c) Polyester
- Q.5 Discuss various Boiler problems and their disadvantages. Give a list of possible preventive measures. 5
- Q.6 Describe EDTA method of determination of hardness of water. 5

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CLASS TEST II

Date: 5th MARCH 2016

B.E. I YEAR

PAPER: CHEMISTRY (CH10502)

Max. Marks: 20

Max. Time: 01 Hr.

Answer ANY TWO questions. All questions carry equal Marks.

- Q.1 Give the advantages and disadvantages of gaseous fuels. Describe Properties, Composition and Applications of :
- Natural gas.
 - Coal gas.
 - Liquified Petroleum gas.
- Q.2 Describe (ANY TWO):
- Structure Property Relationship.
 - Concept of Biodegradation Of Polymers.
 - Synthesis, Properties, Applications of Kevlar and Bakelite.
- Q.3 What do you understand by Viscosity & Viscosity Index of Lubricants? A lubricating oil has a S.U.V. Of 58 seconds at 210°F and of 600 seconds at 100°F. The high viscosity standard (Pennsylvanian) oil has S.U.V. Of 58 seconds at 210°F and 400 seconds at 100°F. The low viscosity index standard (Gulf) oil has a S.U.V. Of 58 seconds at 210°F and 800 seconds at 100°F. Calculate the Viscosity Index of the oil.
- Q.4 Write short notes on (ANY TWO):
- Aniline Point
 - Flash Point & Fire Point
 - Solid Lubricants
- Q.5 Describe the manufacture of Portland Cement by Rotary Kiln method giving a labeled diagram and chemical reactions taking place in various temperature zones.
- Q.6 Write short notes (ANY TWO):
- Gross & Net Calorific Value
 - Cracking
 - Proximate analysis of coal

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CLASS TEST I

Date: 21st JAN. 2016

B.E. I YEAR

PAPER: CHEMISTRY (CH10502)

Max. Marks: 20

Max. Time: 01 Hr.

~~Do not attempt any questions out of the following.~~

Q.1 Define polymers. Describe various classifications of polymers (at least four). Discuss methods of polymerization by taking suitable examples. 10

Q.2 Write short notes on (ANY TWO): 10

- a) Pulverization of coal.
- b) Proximate analysis of coal.
- c) Low temperature and high temperature carbonization of coal.

Q.3 What do you mean by hardness of water? Discuss various types of water hardness. Calculate the temporary hardness, permanent hardness and total hardness (in mg/L, ppm, $^{\circ}\text{Cl}$, $^{\circ}\text{Fr}$) in a given water sample which on analysis gave the following concentration of hardness producing salts: 10

$\text{Ca}(\text{HCO}_3)_2 = 16.2 \text{ mg/L};$

$\text{Mg}(\text{HCO}_3)_2 = 0.0073 \text{ g/L};$

$\text{MgCl}_2 = 9.5 \text{ mg/L};$

$\text{CaSO}_4 = 13.6 \text{ mg/L}.$

Q.4 Give the criteria for selection of good fuel: coal or metallurgical coke. Describe the bomb calorimeter method for determination of calorific value with labeled diagram. 10

Q.5 Discuss following boiler troubles: 10

- a) Scales and sludges.
- b) Boiler corrosion.
- c) Caustic embrittlement.
- d) Priming and foaming.

SHRI SG INSTITUTE OF TECHNOLOGY AND SCIENCE
II Mid Term-Test 2011 CH-1058 (Section-E)

Time-1 Hour

Attempt any four questions. All carry equal marks.

- Q.1 Giving a suitable example explain Electrochemical Theory Of Corrosion.
Q.2 Discuss the Zeolite Process for the removal of hardness of water.
Q.3 What are the essential requirements of an explosive?
Q.4 Explain how corrosion can be minimized by the application of protective coatings.
Q.5 Write short note on- (ANY TWO)
a) Aniline Point of an oil
b) Cutting Fluids
c) Acid Number of an Oil

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B.E. I YEAR, Class Test: I, February, 2015

CH:1058 (Chemistry of Engineering Materials and Processes)

Max. Marks: 15

Time: 01 Hour

Note: Question no. 1 & 2 is compulsory. Attempt any one question from rest question

- Q.1 Discuss hardness of water. How it can be determined with the help of EDTA method? 5

'OR'

What is polymerization? Discuss free radical mechanism of polymerization with suitable example.

- Q.2 Discuss boiler troubles in brief.

'OR'

Write a short note on following:

- a) Nylon-66 b) Bakelite

- Q.3 Draw a neat-labeled diagram of Bomb calorimeter. Write the formula for calculating H.C.V. and L.C.V. 5

- Q.4 Discuss Hydrodynamic mechanism of lubrication. 5

- Q.5 Draw a neat-labeled diagram of Rotary-kiln and discuss chemical reactions taking place in calcination zone. 5

- Q.6 Calculate the amount of lime (84% pure) and soda (92% pure) required for treatment of 20,000 litres of water having the following composition:
Ca(HCO₃)₂ = 40.5 ppm, Mg(HCO₃)₂ = 36.5 ppm, MgSO₄ = 30.0 ppm,
CaSO₄ = 34.0 ppm, CaCl₂ = 27.75 ppm, NaCl = 10.0 ppm. Also calculate the temporary and permanent hardness. 5

NOTE: Attempt ONE question from each section out of A, B, C, D, E

SECTION - A & SECTION - B NOTE: Attempt ANY THREE questions in all.

- Q.1 List the main troubles related to boiler. Explain caustic embrittlement problem in detail.
- Q.2 a) Draw the structural formula of Eriochrome black -T indicator and its structure at pH 8, 10 and 12
b) Draw the labeled diagram of Rotary kiln.
- Q.3 Explain mechanism of free radical polymerization with suitable example.
- Q.4 A water sample contains the following impurities: $\text{Ca}(\text{HCO}_3)_2 = 8.1 \text{ mg/L}$; $\text{Mg}(\text{HCO}_3)_2 = 7.5 \text{ mg/500 ml}$; $\text{CaSO}_4 = 13.6 \text{ mg/2L}$; $\text{MgSO}_4 = 12.0 \text{ mg/L}$. Calculate the total hardness of this water sample in ppm, $^\circ\text{Fr}$ and $^\circ\text{Cl}$

SECTION - C NOTE: Attempt ANY THREE questions in all.

- Q.1 a) Draw the structural formula of Eriochrome black -T indicator and its structure at pH 8, 10 and 12
b) Draw the labeled diagram of Bomb calorimeter.
- Q.2 A water sample contains the following impurities: $\text{Ca}(\text{HCO}_3)_2 = 8.1 \text{ mg/L}$; $\text{Mg}(\text{HCO}_3)_2 = 7.5 \text{ mg/500 ml}$; $\text{CaSO}_4 = 13.6 \text{ mg/2L}$; $\text{MgSO}_4 = 12.0 \text{ mg/L}$. Calculate the total hardness of this water sample in ppm, $^\circ\text{Fr}$ and $^\circ\text{Cl}$
- Q.3 Define gross and net calorific value. Calculate the approximate calorific value by Dulong's formula for a coal sample having following composition: C = 80 %; H = 3.5%; S = 2.8%; O = 5.0 %; N = 1.5% and ash = 7.2%.
- Q.4 List the main troubles related to boiler. Explain caustic embrittlement problem in detail.

SECTION - D NOTE: Attempt ANY THREE questions in all.

- Q.1 Define gross and net calorific value. Calculate the approximate calorific value by Dulong's formula for a coal sample having following composition: C = 80 %; H = 3.5%; S = 2.8%; O = 5.0 %; N = 1.5% and ash = 7.2%.
- Q.2 Describe how the Calorific value of a solid fuel is determined using Bomb calorimeter.
- Q.3 Discuss the estimation of total hardness of a water sample by soap titration method.
- Q.4 What are the 'boiler troubles'? What are the causes of formation of scales & sludges in boiler? Write the disadvantages & ways of prevention.

SECTION - E NOTE: Attempt ANY THREE questions in all.

- Q.1 Draw a neat labelled diagram of Rotary Kiln used in the manufacture of Portland Cement.
- Q.2 Discuss the chemical reactions taking place in different zones of Kiln.
- Q.3 Explain mechanism of free radical polymerization with suitable example.
- Q.4 Discuss the importance of following: (a) Annealing of glass (b) Slaking of lime.
- Q.4 Write note on: (a) Setting and hardening of cement (b) Nylon 6,6

SHRI G. S. INSTITUTE OF TECH. & SCIENCE, INDORE.

Class Test: I, Aug. 2013

B.E. I YEAR

CH1058 (Chemistry of Engineering Materials and Processes)

Max. Marks: 20

Time: 01 Hour.

Note: Attempt any 10 of the questions carrying equal marks.

- Q.1 What do you mean by hardness of water? How is it created? Explain the term degree of hardness.
- Q.2 Explain Caustic embrittlement problem.
- Q.3 A sample of hard water contains following composition: $\text{Mg}(\text{HCO}_3)_2 = 7.3 \text{ mg/L}$, $\text{Ca}(\text{HCO}_3)_2 = 0.0162 \text{ gm/L}$, $\text{MgCl}_2 = 9.5 \text{ mg/L}$, $\text{CaSO}_4 = 13.6 \text{ mg/L}$, then calculate the temporary and permanent hardness of this water sample.
- Q.4 Calculate the amount of lime and soda required for softening 50000 litre of hard water containing following composition: $\text{CaCO}_3 = 25 \text{ ppm}$, $\text{MgCO}_3 = 144 \text{ ppm}$, $\text{CaCl}_2 = 111 \text{ ppm}$, $\text{MgCl}_2 = 95 \text{ ppm}$, $\text{Na}_2\text{SO}_4 = 15 \text{ ppm}$, $\text{Fe}_2\text{O}_3 = 25 \text{ ppm}$.
- Q.5 Give a brief account on Eriochrome black-T indicator's theory applied in EDTA titration.
- Q.6 Explain Free radical type mechanism of polymerization.
- Q.7 Discuss Internal treatment methods for prevention of scale problem.
- Q.8 Give diagram of Rotary kiln and various reactions taking place in it.